Churn scorecard

The "churn scorecard" deliverable has two forms in this project:

1. **Customer scores:** every customer's calibrated churn probability and risk band (scorecard table); the targeting tables add value, offer, expected value and the reason for each decision.
2. **Points card (this page):** the deployed model rewritten as whole points, so a call-centre agent or CRM rule can score a customer without the model and see why the score is high.

How to read it

- Start from **133 points** and add the points for each of the customer's attributes.
- **500 points = 50% churn probability.** Every **40 points doubles the odds** of churning; 40 points fewer halves them.
- Higher means riskier. The points come from the calibrated Logistic Regression (calibration folded in); tenure and monthly charges are banded.
- On the 1,409 test customers the card's probability is within **1.0 percentage points** of the model's on average (largest gap 6.4 points; validation 1.0). It ranks customers as well as the model (ROC-AUC 0.831 vs 0.830).

Points measure churn risk only. Whether to make an offer also depends on the customer's value and the offer's cost; use the targeting list for that.

Points

Characteristic	Attribute	Points
tenure	0-3 months	120
tenure	4-6 months	113
tenure	7-12 months	106
tenure	13-18 months	95
tenure	19-24 months	83
tenure	25-36 months	69
tenure	37-48 months	46
tenure	49-60 months	26
tenure	61+ months	0
MonthlyCharges	up to $30	0
MonthlyCharges	$30-45	14
MonthlyCharges	$45-60	23
MonthlyCharges	$60-75	34
MonthlyCharges	$75-90	43
MonthlyCharges	$90-105	53
MonthlyCharges	over $105	62
Partner	Yes	1
Partner	No	0
Dependents	Yes	0
Dependents	No	17
PhoneService	Yes	0
PhoneService	No	38
PaperlessBilling	Yes	22
PaperlessBilling	No	0
MultipleLines	Yes	15
MultipleLines	No	0
OnlineSecurity	Yes	0
OnlineSecurity	No	25
OnlineBackup	Yes	0
OnlineBackup	No	10

DeviceProtection	Yes	0
DeviceProtection	No	1
TechSupport	Yes	0
TechSupport	No	25
StreamingTV	Yes	6
StreamingTV	No	0
StreamingMovies	Yes	9
StreamingMovies	No	0
InternetService	DSL	30
InternetService	Fiber optic	61
InternetService	No	0
Contract	Month-to-month	75
Contract	One year	34
Contract	Two year	0
PaymentMethod	Electronic check	25
PaymentMethod	Mailed check	8
PaymentMethod	Bank transfer (automatic)	0
PaymentMethod	Credit card (automatic)	0

Score bands on test customers

Points	Customers	Card probability	Model probability	Observed churn
up to 380	552	4.2%	4.3%	4.5%
381-420	173	15.8%	15.9%	16.2%
421-460	192	26.5%	26.8%	26.6%
461-500	178	42.2%	42.3%	44.9%
501-540	185	58.3%	58.3%	52.4%
541-580	115	72.3%	72.1%	70.4%
over 580	14	82.1%	81.7%	85.7%

What drives the points

Odds ratios of the deployed model; each category is compared with its lowest-churn level and each flag with not having the service. These are associations in a snapshot, not causes: a customer who signs a two-year contract does not acquire a two-year customer's risk.

Field	Comparison	Odds ratio
Contract	Month-to-month vs Two year	3.66
InternetService	Fiber optic vs No	2.90
Contract	One year vs Two year	1.79
InternetService	DSL vs No	1.67
PaymentMethod	Electronic check vs Credit card (automatic)	1.53
PaperlessBilling	Yes vs No	1.46
MultipleLines	Yes vs No	1.29
StreamingMovies	Yes vs No	1.17
PaymentMethod	Mailed check vs Credit card (automatic)	1.15
MonthlyCharges	+10 USD a month	1.13
StreamingTV	Yes vs No	1.10

Partner	Yes vs No	1.01
PaymentMethod	Bank transfer (automatic) vs Credit card (automatic)	1.01
DeviceProtection	Yes vs No	0.98
OnlineBackup	Yes vs No	0.84
Dependents	Yes vs No	0.74
tenure	+12 months	0.69
TechSupport	Yes vs No	0.65
OnlineSecurity	Yes vs No	0.65
PhoneService	Yes vs No	0.52

Use and limits

- Recompute the card whenever the model is retrained; points from different runs are not comparable.
- Do not add age or gender: they are not model inputs and are audited separately (model card).
- The points are fitted on the IBM sample; a real operator needs its own card (data readiness).